YASHWANTRAO CHAVAN MAHARASHTRA OPEN UNIVERSITY, NASHIK

(Estd. By Government of Maharashtra)
Dnyangangotri, Near Gangapur Dam, Govardhan, Nashik-422 222
Home Assignment

Academic Year - 2025 - 2026

Name of the Programme & Prog. Code	**P131 - B.C.A.(with Credits) - Distance Education - 2016 CGPA - S.Y.B.C.A Semester III & P131**		
Year / Semester	**S.Y.B.C.A / Semester III**		
Name of the Course & Course Code	**Python Programming (ICT152)**		
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Name of the Study Centre & Code	**Zagdu Singh Charitable Trust's Thakur Shyamnarayan Degree College & 31509**		
Date of Submission	14/12/25	**No of Pages**	8

General Instructions

1	Home assignment is useful for self-analysis of your preparation of the final examination and progress
2	Read all the questions and their marks in the home assignment carefully understand the Definitions and concepts properly
3	Read carefully the text, syllabus and summary related to home assignment.
4	Do not copy the points, matter from the text while writing home assignment mentions your one point /opinions whenever necessary
5	Your answer should include how you would apply the knowledge in real life situations
6	Use A4 size lined papers for writing home assignment by taking printout from this file only
7	Write each new answer on a different page.
8	Printed or typed answers are not allowed
9	The marks of Home assignment and corrective instructions will be sent after checking home assignment
10	Present the Home Assignment by following the instructions and rules. Read carefully all the instructions and rules before any correspondence or communication with the university
11	**For the programs under NEP 2020 Pattern each theory course of 4 credits its carry 30 marks, there are Two Home Assignments:** i) **Home Assignment 01 (HA01) is of Twenty marks** consisting of 2 Short Answer Questions carrying 5 marks each and 1 Long Answer Question of 10 marks. ii) **Home Assignment 02 (HA02) is of Ten marks** consisting of total 10 Objective Type Questions (MCQ) carrying 1 mark each.
12	**For each theory course of 2 credits, there is Only One Home Assignment of Fifteen Marks** consisting of 1 Short Answer Question of 5 marks and 1 Long Answer Question of 10 marks.
13	Student have to write the Home assignment 02 (MCQ) on separate page. Home Assignment 02 also to be uploaded along with Home assignment 01.

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Q.NO.

1. Explain the usage of try and except in Python Programming.

→ In Python programming, try and except blocks are used to handle runtime errors, which are also known as exceptions. An exception is an unexpected event that occurs. An during the execution of a program and disrupts its normal flow. Instead of stopping the program abruptly, Python provides the try-except mechanisms to detect errors and handle them gracefully. This makes the program more reliable, user-friendly and easier to debug.

The try block is used to write the code that may generate an error. Python execute this block first. If except block is the program continues normally and the except block is skipped. However, if an error occurs, Python immediately transfers control to the except block, where the errors is caught and handled. The except block contains the code that defines what action should be taken when a particular error occurs.

Using try and except improves the robustness of a program because it prevents sudden termination and allows the programmer to display meaningful error messages instead of technical crash reports. It also helps in handling common errors such as division by zero, invalid user input, or file handling issues. In Python, different types of exceptions can be handled using different except blocks. For example, ZeroDivisionError, ValueError and TypeError are commonly used exception types. A general except block can also be used to handle any type of error if the exact exception is not known.

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Q.NO. 1 continuing

Syntax of try and except in Python:

```
try:
    # code that may cause an error
except ErrorType:
    # code that handles the error
```

Here, the try block contains risky code, and the except block specifies the type of error and the response to that error. If the mentioned error occurs, the except block is executed.

Example:

consider a situation where the users enters two numbers and the program performs division. If the user enters zero as denominator, the program will generate an error. Using try and except, this error can be handled properly.

```
try:
    result = a/b
except ZeroDivisionError:
    print("Division by zero is not allowed.")
```

In this example, the division operation is placed inside the try block. If the denominator becomes zero, Python raises a ZeroDivisionError, which is then caught by the except block, and a friendly message is displayed instead of crashing the program. Thus, the usage of try and except in Python Programming allows developers to identify potential errors, handle them efficiently, and ensures smooth execution of programs.

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Q.NO.

2. What is Array slicing? How can it be used to access one-dimensional and multidimensional sub arrays?

→ Array slicing in Python is a powerful technique used to extract a specific portion of an array, known as sub array, from an existing array. Instead of accessing elements one by one, slicing allows the programmer to retrieve a group of elements in a single operation. This technique is mainly applied to one-dimensional arrays and multidimensional arrays, which are commonly represented using lists or arrays-based libraries in Python.
In simple terms, array slicing means selecting a continuous range of elements from an array by specifying starting index, ending index, and step value. Python follows zero-based indexing, meaning the index of the first element is always zero. Array slicing does not modify the original array; instead, it creates a new sub array containing the selected arrays.

Array Slicing in One-Dimensional Arrays:
A one-dimensional array is a linear collection of elements stored in a single sequence. In such arrays, slicing is used to access a subset of elements based on their positions. This is especially useful when working with large datasets where only a specific range of values is required.
Syntax:

```
array[start : end : step]
```



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Q.NO.

4 Explain how classes and objects are created in Python with example.

→ In Python programming, classes and objects are the core concepts of object-oriented Programming (OOP). OOP allows programs to be written in a structured, reusable, and real-world-oriented manner. A class acts as a blueprint for creating objects while an objects is a real instance of that class.

A class is a user-defined data type that groups data members (variables) and member function (methods) into a single unit. It defines the properties and behavior that the object created from it will have. On the other hand, an Object represents a real-world entity and occupies memory. Each object created from the same class can have different values for its data members.

In Python, classes are created using the class keyword. Inside a class, variables store data, and function define operations that can be performed on that data. The special method __init__() is called a constructor, and it is automatically executed when an object of the class is created. It is mainly used to initialize the data members of the class.

Syntax for creating a class in Python:

```
class className:
    def __init__(self, Parameters):
        #initialization of data members.
    def method_name(self):
        # method definition.
```

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Q.NO.

3. How to use Python's widgets ? Explain functionality of any one widget.

→ In Python, widgets are graphical user interface (GUI) components that are used to interact with the user. widgets allows users to input data, display information, and perform actions by clicking buttons, selecting options, or entering text. Python widgets are commonly used in GUI libraries such as Tkinter, Pyqt, and kivy, with Tkinter being the most widely used and built-in GUI library in Python. Python widgets make programs more interactive and user-friendly by providing visual interface instead of relying only on command-line input. widgets help developers create desktop applications like calculators, forms, login screens, and data entry system.

To use Python widgets, the following general steps are followed:
1) Import the required GUI library (such as Tkinter).
2) Create the main application window.
3) Create widgets like lables, buttons, text boxes, etc
4) Place the widgets using layout managers.
5) Run the applications loop to display the window.

Functionality of the Button widget

The Button widget is one of the most commonly used widgets in Python GUI programming. It is mainly used to perform an action when the user clicks on it. Buttons are essential components of interactive applications because they allow users to control program flow easily. The primary

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Q.NO. 4 Continuing

To create an object, the class name is used like a function call. Each object can access the variables and methods defined inside the class using the dot (.) operator.

Example of class and object in Python:

```
class Student:
    def __init__(self, name, roll_no):
        self.name = name
        self.roll_no = roll_no
    def display_details(self):
        print("Name:", self.name)
        print("Roll Number:", self.roll_no)

# Creating object of the class
S1 = Student("Rahul", 101)

# Accessing class method using object
S1.display_details()
```

Explaination of the Example:

In this example, a class named Student is defined using the class keyword. The __init__() method initialize the student's name and roll number whenever a new object is created. The keyword self refers to the current object and is used to access the class variables. The method display_details() is used to display the student information. After defining the class, an object S1 is created using the class name Student. The object stores actual values and can access the methods of the class. When S1.display_details() is called, the function prints the data.

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Q.NO. 3 continuing

function of a Button widget is to execute a special task or function when it is clicked. For example, submitting a form, calculating result, or closing an applications are commonly handed by buttons.

Definition:

A Button widget is a GUI element that represents a clickable button on the screen and executes a predefined function or command when pressed.

Syntax of Button widget in Python (Tkinter):

```
button=Button(parent_window, text="ButtonName",
              command=function_name)
```

Here,

- parent_window refers to the main window or frame
- task specifies the label displayed on the button,
- command defines the function to be executed when the button is clicked.

The Button widget responds to user actions such as mouse clicks and provides immediate feedback by performing the assigned task. This event-driven behaviour makes the Python GUI applications dynamic.

Working Explanations:

When a Button widget is created and displayed on the screen, it waits for a user actions. Once the user clicks the button, Python automatically calls the function associated with the command parameter. This function may perform calculations, update text, display messages, or trigger other widgets. In conclusion, Python widgets are essential tools for building interactive graphical applications.

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Q.NO. 2 continuing

Here,

- Start represents the index where slicing begins.
- end represents the index where slicing ends.
- Step determines the interval between elements.

If the start value is omitted, slicing begins from the first element. If the end value is omitted, slicing continues till the last element. Using this syntax, a one-dimensional array containing a specific range of values can be easily accessed.

Array Slicing in Multidimensional Arrays:

A multidimensional array consists of multiple one-dimensional arrays arranged in rows and columns, similar to a matrix. In Python, this is usually represented using a lists of lists. slicing in multidimensional arrays allows access to specific rows, columns, or blocks of data, forming multidimensional subarrays. To access these elements, indexing is applied for each dimension. For example, the first index refers to the row, and the second index refers to the column. Slicing can be applied at once at one or more dimensions to extract sub arrays.

Syntax:

array[row_index][column_index]

To access sub arrays slicing can be used along rows and columns individually. This technique enables retriving:

- A complete row as a sub array.
- A complete column as a sub array
- A specific block or region from array.

when using libraries, like Numpy, slicing becomes efficien

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